Data-driven Investigations of Culinary Patterns in Traditional Recipes across the World

Navjot Singh[1,2] & Ganesh Bagler[1]

[1]*Center for Computational Biology, Indraprastha Institute of Information Technology, Delhi*

[2]*Delhi Technological University*

Prologue

The knack for combining and processing raw ingredients has been suggested to be critical for the evolution of large brains.

The penchant for cooking has led to diverse cooking styles across cultures which are crystallized as elaborate procedures in traditional recipes.

Passed down the generations, these recipes form the core of **cultural identity** and shape the dietary habits.

Motivation

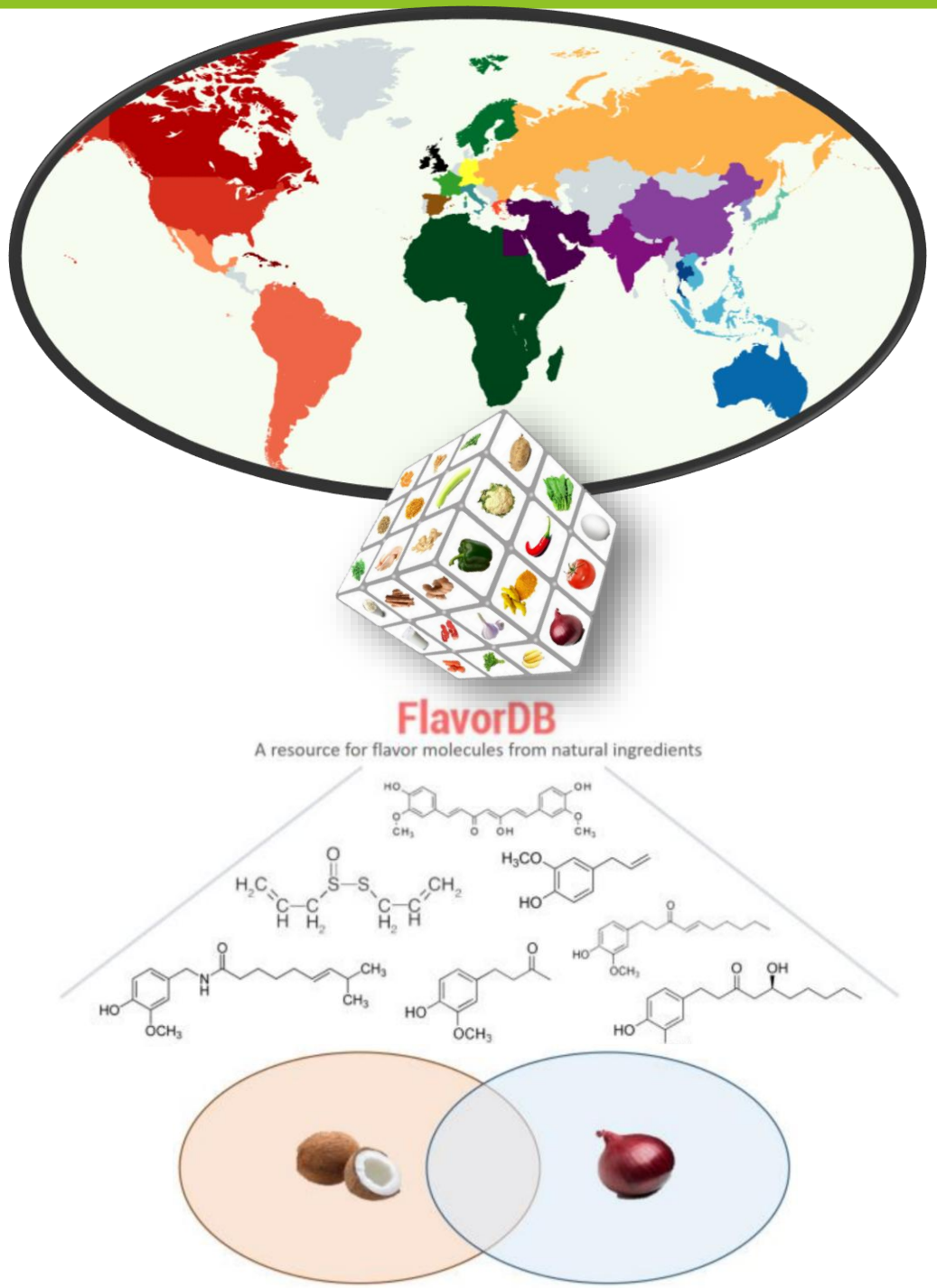
While it is clear that preparation and perception of food are complex phenomena that make it difficult to determine underlying rules, search for patterns in recipes can provide much-needed impetus in **developing the scientific foundation**.

Analysis of data pertaining to recipes, ingredients and relevant features can help **uncovering underlying patterns** in traditional recipes, paving way for innovation in food design and flavor pairings.

Bird's Eye View

We present a framework for multi-level investigation of traditional recipes rooted in the annotated data of **recipes** from diverse world cuisines and those of **flavor molecules** from natural **ingredients**.

Systematic investigation of deviation patterns from random compositions provide insights into 'culinary fingerprints' that can form the basis for synthesis of novel recipes as well as targeted alterations in existing recipes.



Data of Recipes

Compilation



16177





2609



45772 Recipes



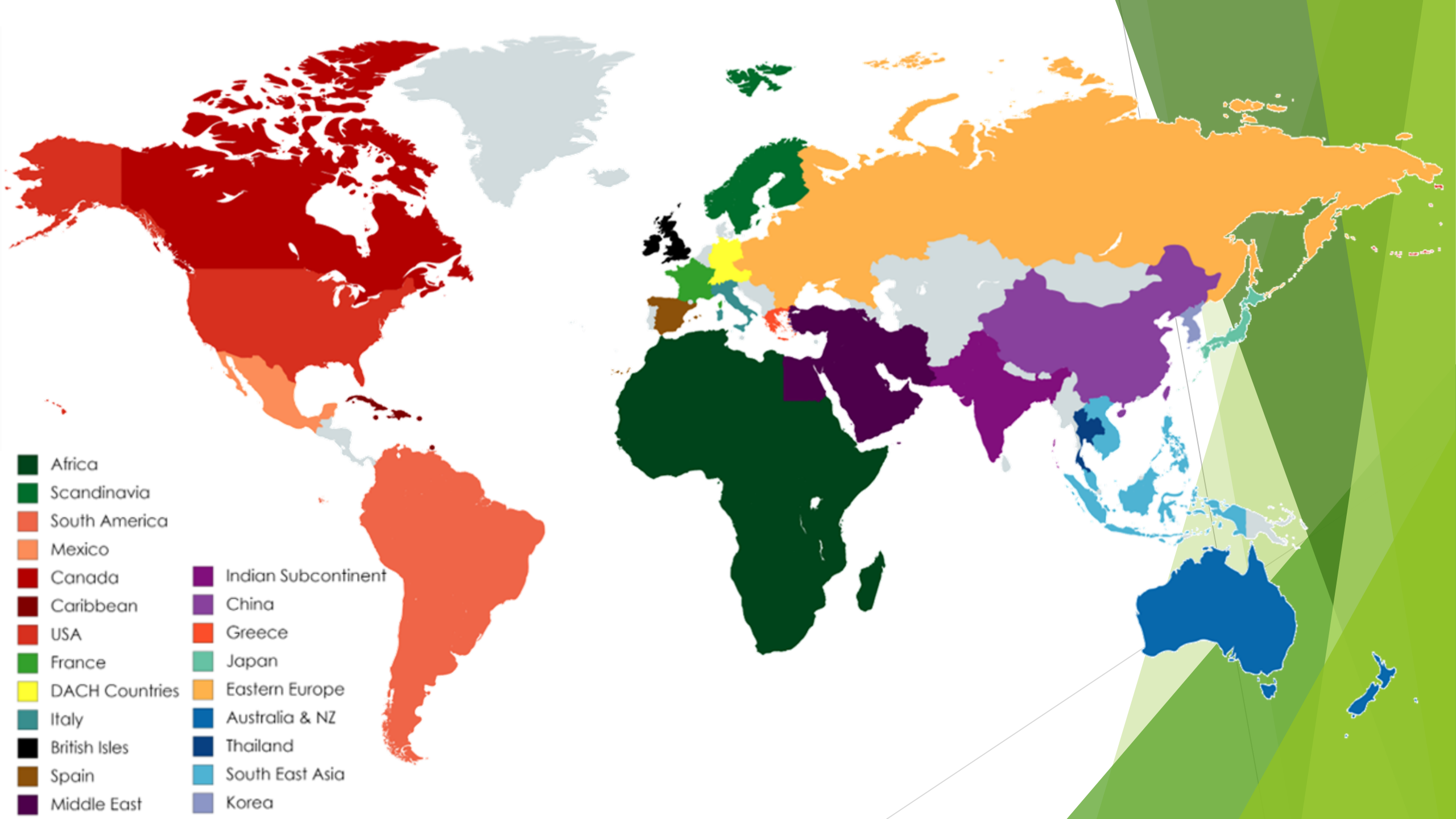


15917





11069


Africa
Scandinavia
South America
Mexico
Canada
Caribbean
USA
France
DACH Countries
Italy
British Isles
Spain
Middle East
Indian Subcontinent
China
Greece
Japan
Eastern Europe
Australia & NZ
Thailand
South East Asia
Korea

Statistics

Region	Recipes	Ingredients
Africa (AFR)	651	303
Australia & New Zealand (ANZ)	494	294
British Isles (BRI)	1075	340
Canada (CAN)	1112	368
Caribbean (CBN)	1103	340
China (CHN)	941	302
DACH Countries (DACH)	487	260
Eastern Europe (EE)	565	255
France (FRA)	2703	424
Greece (GRC)	934	280
Indian Subcontinent (INSC)	4058	378

Region	Recipes	Ingredients
Italy (ITA)	7504	452
Japan (JPN)	580	283
Korea (KOR)	301	198
Mexico (MEX)	3138	376
Middle East (ME)	993	313
Scandinavia (SCND)	404	245
South America (SAM)	310	221
South East Asia (SEA)	611	266
Spain (ESP)	816	312
Thailand (THA)	667	265
USA (USA)	16118	612

Data of Flavor Molecules



FlavorDB comprises of 25595 flavor molecules among which 2254 are associated with 936 natural ingredients. However, this list was tailored for the analysis of recipes:

- 29 generic and noisy entities were removed.
- For certain ingredients, their common names were added.
- Linguistic and spelling variations of ingredients were accounted for.
- 13 new ingredients for which flavor profiles were available but were coarse-grained in FlavorDB were added due to their role in recipes.
- Data from Ahn et al., was used to include 4 more unique ingredients.
- Information for 7 additives was manually added considering their high usage.

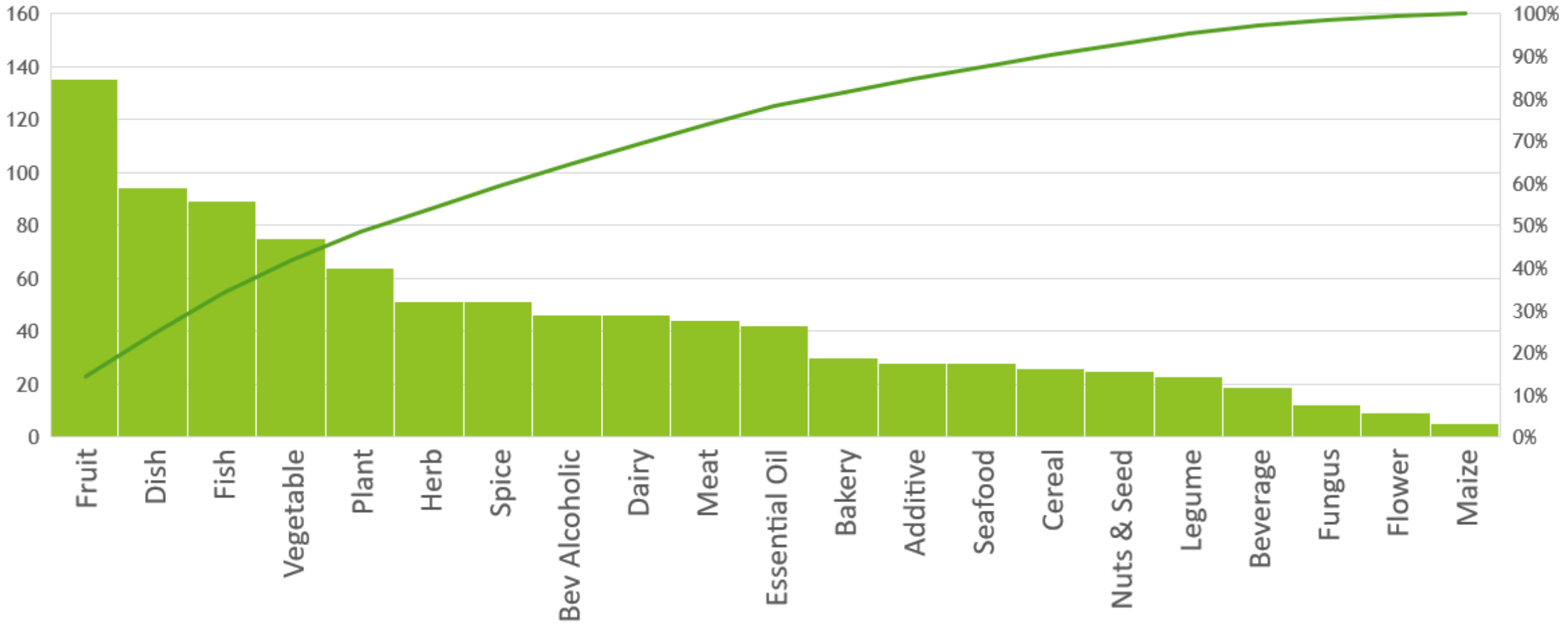
Compound Ingredients

In addition to 840 unique basic ingredients, we compiled a list of 103 'compound ingredients'. This list includes:

- Readymade spice combinations, sauces and other commonly used dishes which are often considered as core ingredients themselves.
- Some entities from FlavorDB were bundled to create a compound ingredient to account for limited number of flavor molecules in certain ingredients.

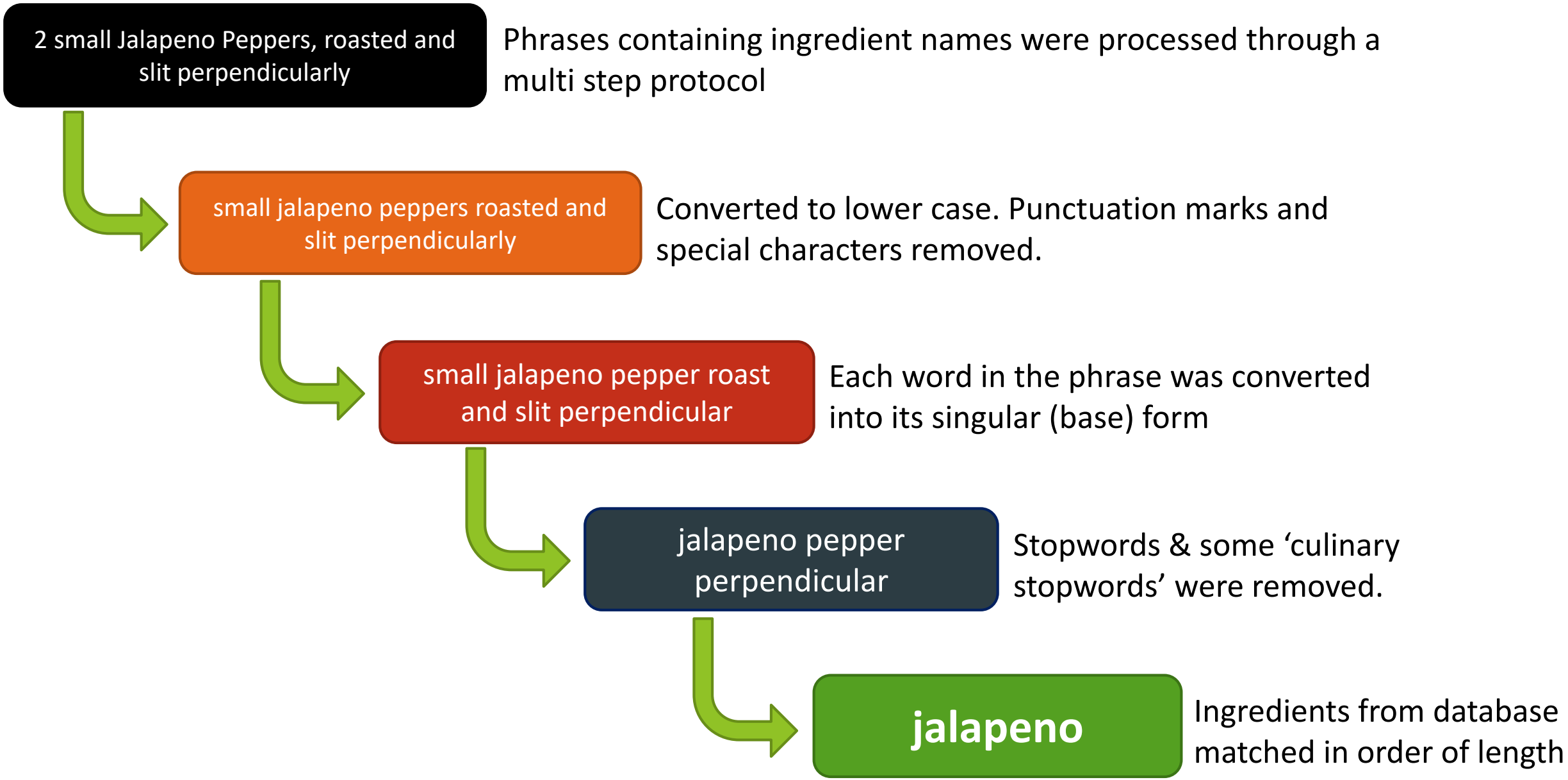
Recategorization

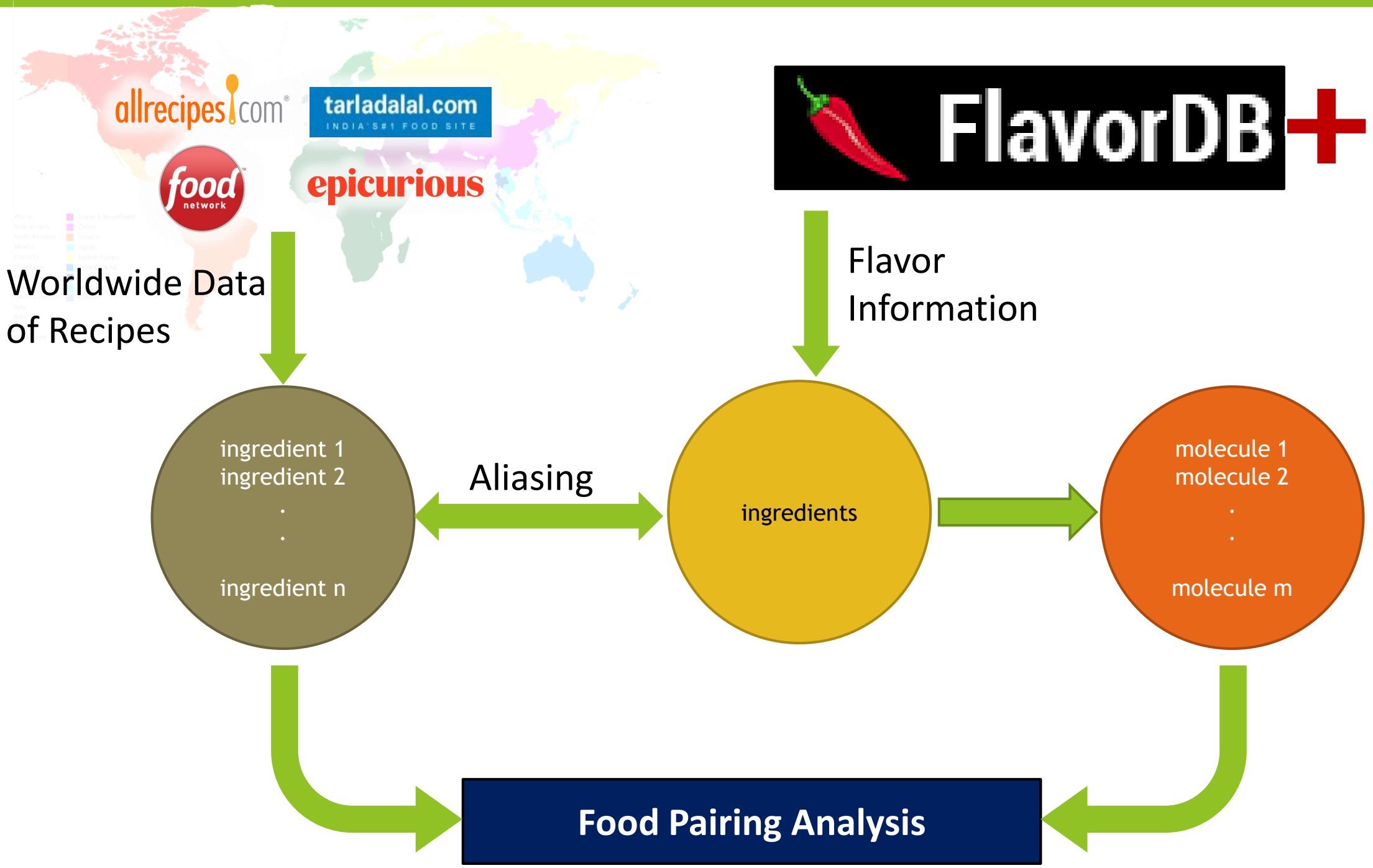
FlavorDB's ingredient categorization was inherited, however recategorization was done to reduce number of categories. Therefore, eventually each of the 942 ingredients was classified into one of the 21 categories.



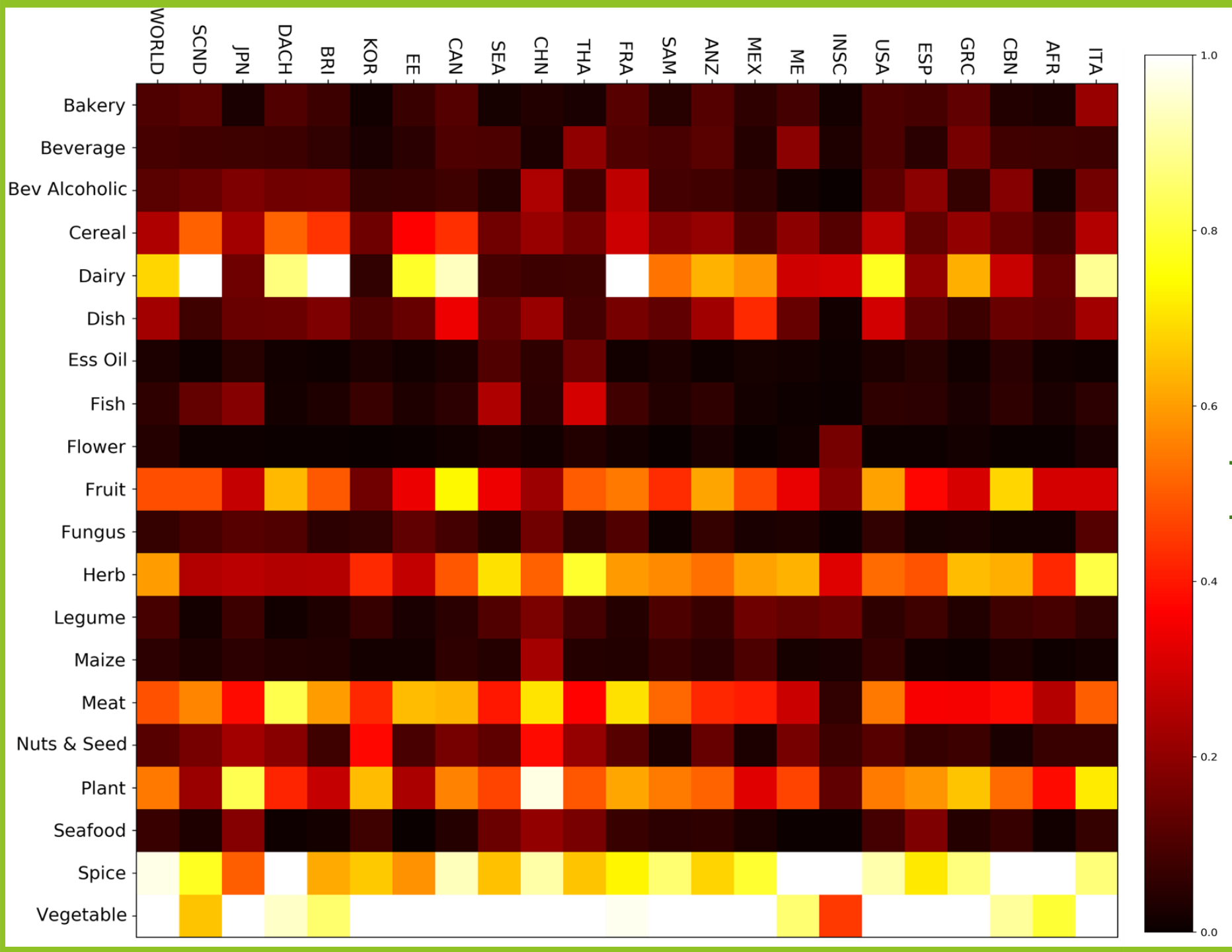
Bringing it all together

Aliasing Protocol



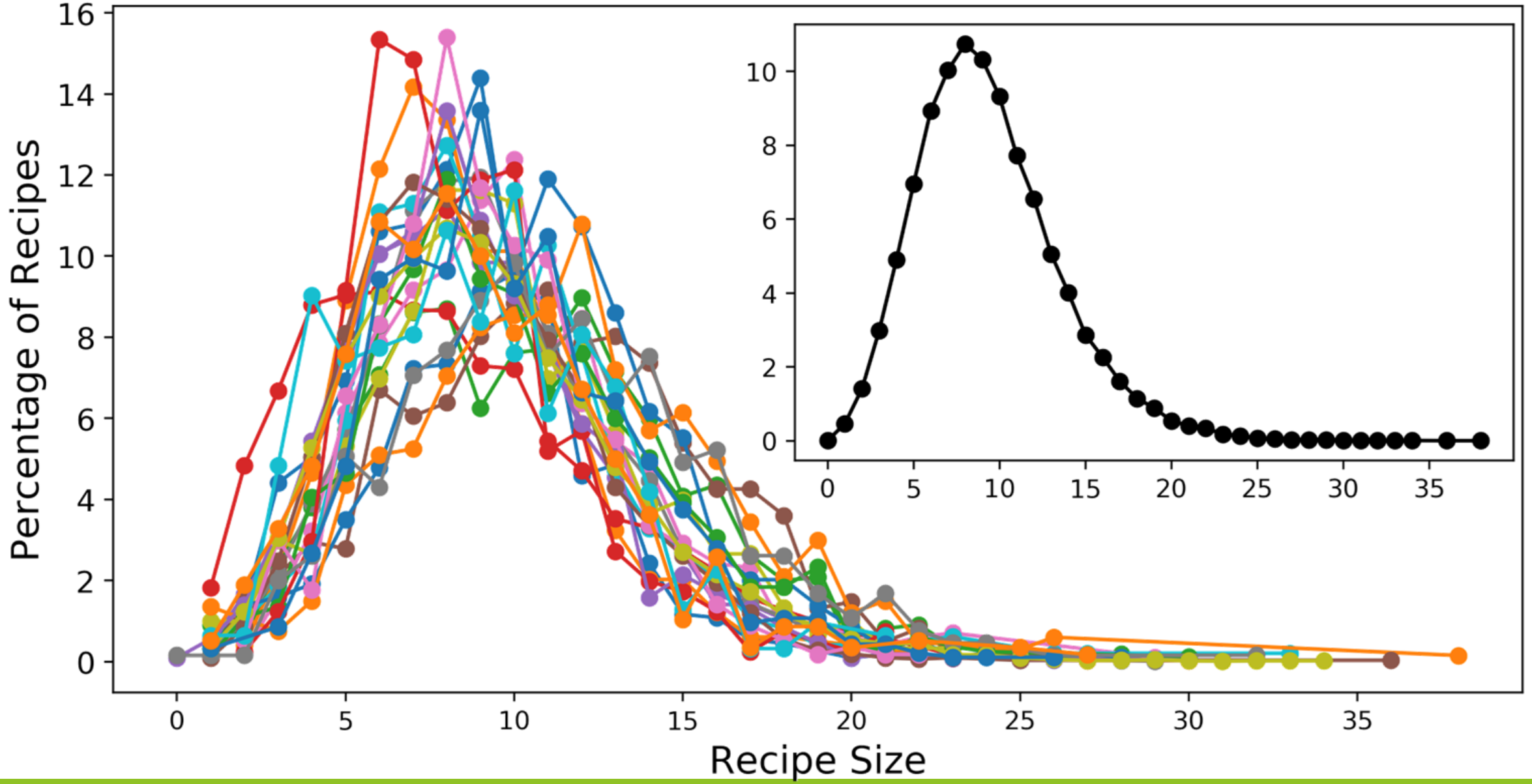

allrecipes.com
tarladalal.com
INDIA'S#1 FOOD SITE
food network
epicurious
FlavorDB+
Worldwide Data of Recipes
Flavor Information
ingredient 1
ingredient 2
.
.
ingredient n
Aliasing
ingredients
molecule 1
molecule 2
.
.
molecule m
Food Pairing Analysis

Data Analytics

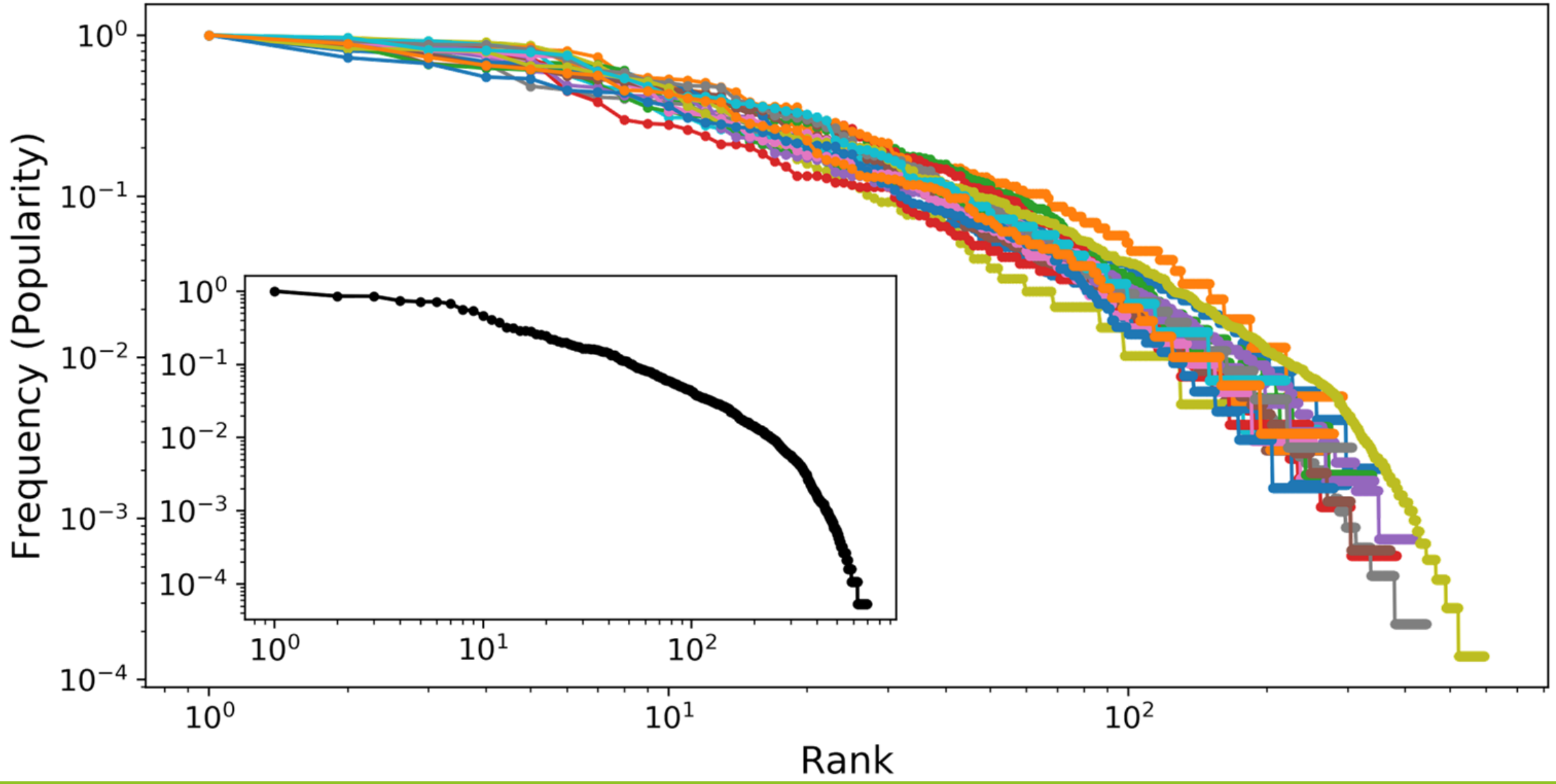


Category Importance Heatmap

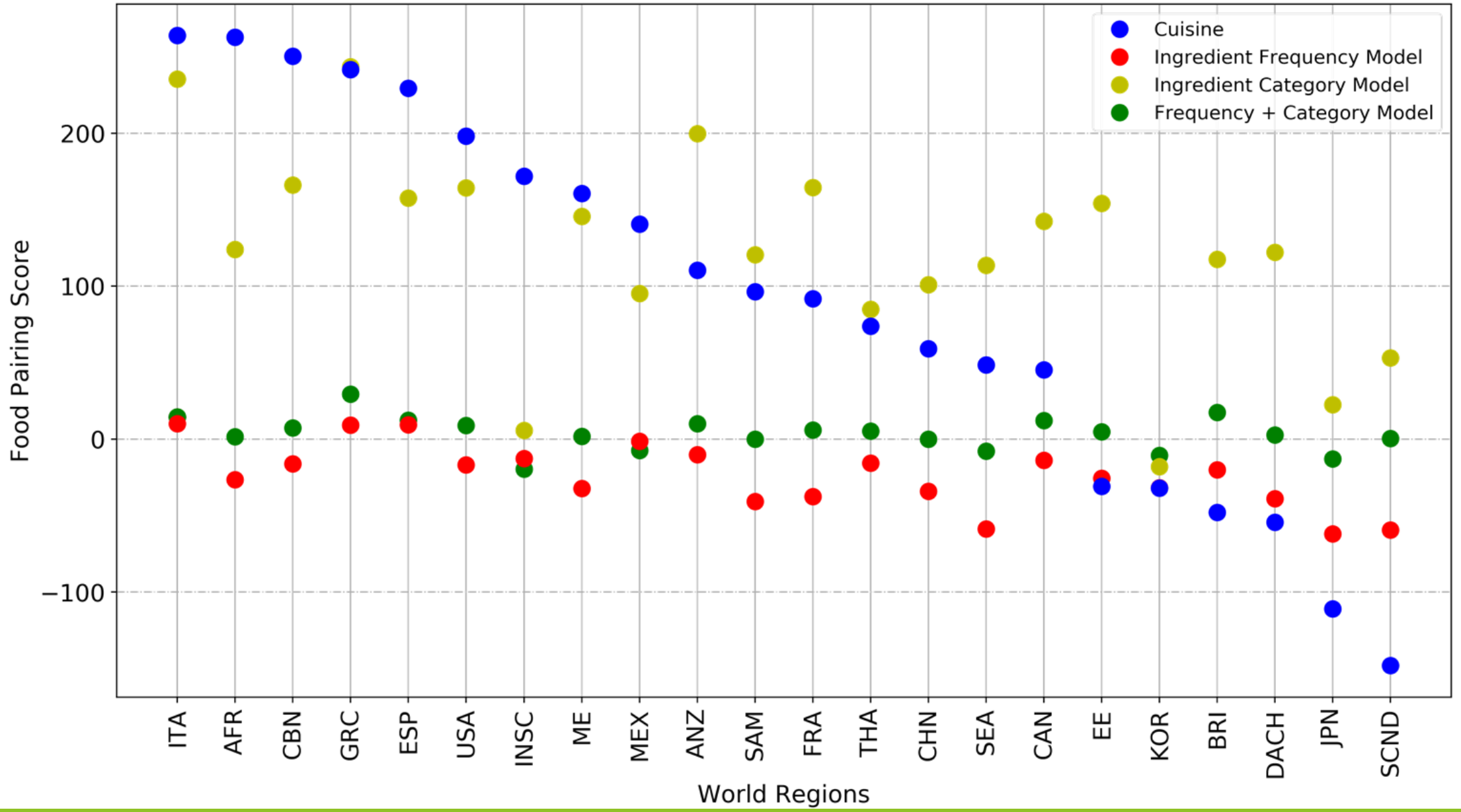
Recipe Size Distribution



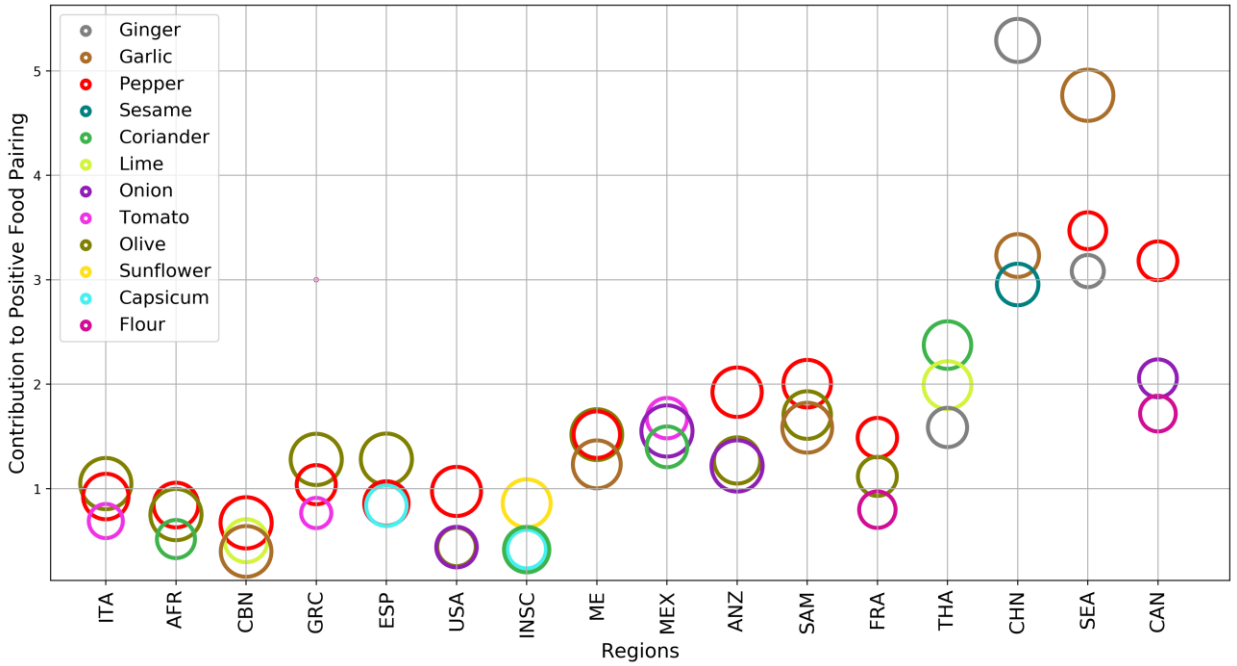
Ingredient Popularity Distribution

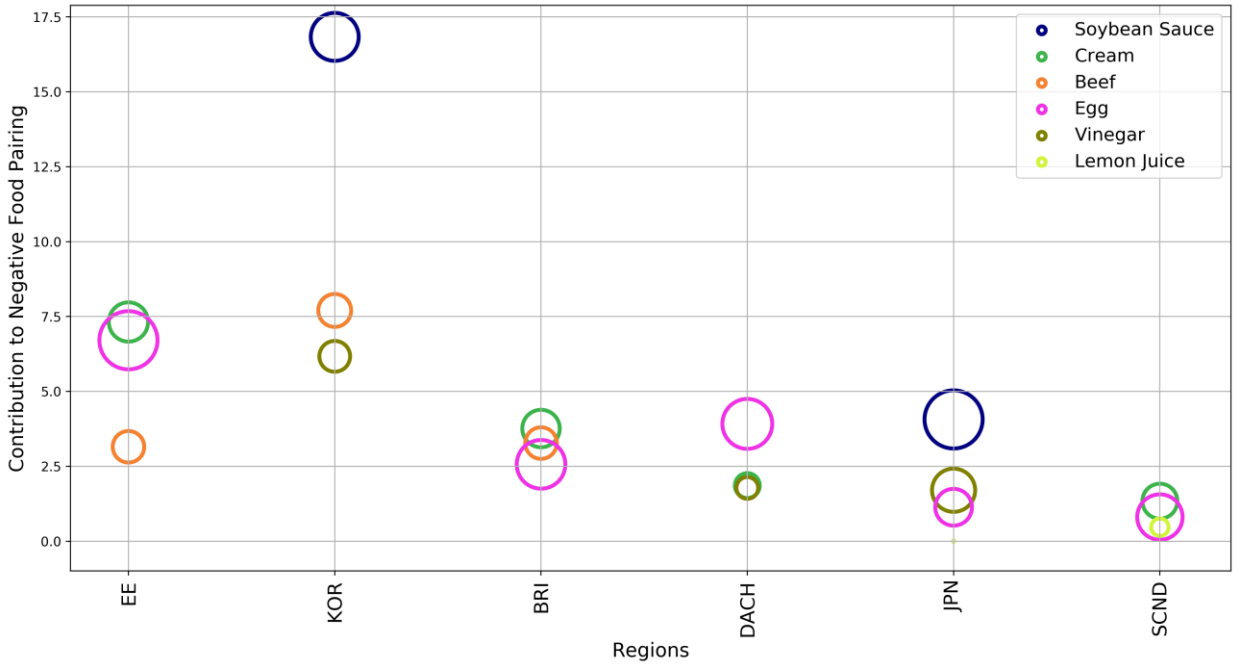


Flavor Pairing



Ingredient Contribution





Shortcomings

- Recipes attributed to a particular cuisine at any of the online source(s) might not be “traditional” per se.
- While there is a large number of ingredients in the database, it is far from exhaustive.
- Ingredients not identified by aliasing protocol are labelled for manual curation.
- Same recipe might be submitted at multiple sources, no real way to eliminate redundancies.

Published

‘Data-driven investigations of culinary patterns in traditional recipes across the world’, Navjot Singh and Ganesh Bagler* Accepted for presentation and publication at the 1st edition of the DECOR 2018 @ IEEE International Conference on Data Engineering (ICDE), to be held in Paris from 16-19 April 2018.

Preprint available on ArXiv: https://arxiv.org/abs/1803.04343